S3 Supplementary: Embodied Anchoring Principle (EAP)

Formal integration with RVC, neural substrates and rhythms, exploratory phenomenology, the measurement plan, and empirical status from companion analyses

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Relation to the main article

This supplement expands the **Embodied Anchoring Principle (EAP)** appendix pointer from the main article (`NoeticDiffusionTheory.typ`, Appendix). EAP is a **Tier-2** extension: its predictions target a later wave of interoception-focused and neurophenomenological studies. The core Tier-1 tests concern the minimal MNDM, functional RVC, and MNPS geometry, and do not require any result reported here.

S3.0 Role and empirical priority

EAP states that denoising is **conditioned** on interoceptive/body state: A **self-prior** anchors reconstruction to a world-for-me. The anterior insula (AIC) and ACC adjust precision on interoceptive channels and bind interoceptive with exteroceptive streams, thereby shaping which latent samples are stabilized and **when** (Critchley & Harrison, 2013; Khalsa et al., 2018; Wang et al., 2019). In terms of empirical priority, EAP should be read as a **Tier-2** extension of NDT: its predictions are intended for a later wave of interoception-focused and neurophenomenological studies, whereas the core Tier-1 tests concern the minimal MNDM, functional RVC, and MNPS geometry.

S3.1 Model implementation and integration with RVC

Formal integration. EAP introduces two forms of embodied modulation: (i) landscape conditioning via a self-prior, and (ii) variance schedule entrainment via bodily rhythms. We specify both mathematically.

Landscape conditioning.

$$F(X,t) \mapsto F(X,t) + \lambda(t) F_{\text{self}}(X, s_t), \quad \alpha, \beta = \alpha_0 + \kappa_\alpha h(s_t), \beta_0 + \kappa_\beta h(s_t) \tag{1}$$

where:

- $s_t = \big(\varphi_{\text{cardiac}}(t), \varphi_{\text{resp}}(t), A_{\text{HEP}}(t)\big)$ is the interoceptive state vector (cardiac phase, respiratory phase, HEP amplitude).
- $F_{\text{self}}(X, s_t)$ penalizes neural states X that are incongruent with bodily state s_t (for example, high-arousal neural signatures during physiological calm).
- $\lambda(t)$ sets self-prior strength, decomposed as $\lambda(t) = \lambda_{\text{signal}}(t)\lambda_{\text{binding}}(t)$, where $\lambda_{\text{signal}} \propto A_{\text{HEP}}$ (interoceptive signal strength) and $\lambda_{\text{binding}} \propto \text{FC}_{\text{AIC-PCC}}$ (ownership binding).
- $h(s_t)$ is a context-sensitive modulator that adjusts landscape weights.

Alternative gating formulation. While Eq. 1 presents EAP as an additive potential term (energetic cost), it may also be implemented as a **gain control** or gating mechanism, where $\lambda(t)$ modulates the **slope** of F (steepening gradients in task-relevant directions) rather than adding a separate basin. Both formulations predict that body state shapes the attractor landscape, but the gating model emphasizes signal-to-noise ratio modulation.

Identifiability note. Because both the $d(X)$ term in the extended potential and the self-prior F_{self} penalize "misconfigured" states, there is a risk of confusing generic deviations from optimal metastability with specifically embodied misalignment. Our empirical strategy is to **isolate** F_{self} using paradigms with strong interoceptive manipulation (for example, cardio-visual synchrony, threat vs. safety contexts) and to examine whether EAP-related indices (HEP, AIC–PCC connectivity, phase-coupling terms in $\lambda(t)$) explain variance in phenomenology and MNPS geometry beyond what is captured by $d(X)$ alone. In early datasets, we therefore treat EAP effects as exploratory and require that any putative self-prior signatures remain after controlling for $d(X)$-based metastability deviations.

Variance schedule entrainment.

$$r(t) \mapsto r(t)\big[1 + \gamma_{\text{entrain}} \cos(\varphi_{\text{cardiac}} - \varphi_{\text{theta}}) \cos\big(\varphi_{\text{resp}} - \varphi_{\text{alpha}}\big)\big] \tag{2}$$

This is a **formal target**, not an estimator used in the present article or a calibrated biological law. If such coupling exists, phase alignment could modulate the effective schedule; its strength, sign, and state dependence remain empirical questions. We therefore do not assign a fixed empirical value to γ_{entrain}.

S3.2 Empirical status from companion analyses

The current implemented measurement layer is AnchorState: windowed HRV, PPG, and pupil-derived proxies such as `vagal_index`, `sympathetic_index`, `vascular_index`, and `anchor_index`. These are operational autonomic/arousal proxies, not direct measurements of HEP, vagal nerve activity, AIC–PCC binding, or $\lambda_{\text{signal}}(t)$. In a working-memory dataset, the anchor surface and MNPS/trajectory geometry varied non-monotonically with task phase, whereas direct block-native anchor–MNPS coupling was modest and between-subject correlations were null after correction. This supports a constrained condition-level co-modulation result, not causal body-to-manifold gating.

In the RichSleep phase-resolved data, autonomic phase locking was carrier-specific: slow-oscillation trough/upstate timing was stronger than spindle-onset timing, and REM-theta did not provide a uniform positive result. The spindle-onset conclusion was method-sensitive across null constructions, and intrinsic event strength did not yield FDR-supported alignment-to-geometry efficacy. Thus event–body timing is EAP-relevant but does not establish Eq. 2.

S3.3 Metabolic perturbation boundary (not EAP validation)

A paired type-1-diabetes clamp EEG reanalysis provides a useful counterexample to treating slow rhythmic structure as a scalar benefit. Hypoglycemia was associated with lower 3D mobility and a mixed stratified signature (lower m_a, m_e, and e_e; higher d_n and e_m), while neither prespecified 3D Jacobian endpoints nor reachability log-volume supported a confirmed reachability collapse after the relevant correction. The dataset contains no continuous glucose, recovery, cardiac, respiratory, HEP, interoceptive-awareness, or behaviour measures. It is therefore a bounded metabolic condition contrast, not a test of EAP, a metabolic adequacy factor, or a consciousness measure; it is also unrelated to HELS landscape learning.

Mechanistic pathway (AIC → variance modulation).

$$\text{Cardiac signal} \underset{\text{NTS/insula}}{\rightarrow} A_{\text{HEP}} \underset{\text{AIC}\,\rightarrow\,\text{L6 CT}}{\rightarrow} \text{Corticothalamic gain} \underset{\text{L6}\,\rightarrow\,\text{TRN}}{\rightarrow} r(t) \;\; \text{modulation} \tag{3}$$

AIC detection of interoceptive signals modulates layer-6 corticothalamic (CT) neuron excitability via local interneuron circuits (Wang et al., 2019). Since L6 CT neurons control TRN firing (see the main article's Rhythmic Variance Control and TRN gating section), interoceptive state indirectly gates variance: Strong interoceptive signal → enhanced L6→TRN drive → tighter variance clamping. This creates the observed cardiac-phase-dependent perceptual gating (Salomon et al., 2016).

S3.4 Neural substrates and rhythms

Insula/ACC as precision and binding. AIC adjusts gain on interoceptive channels (precision control) and participates in interoceptive–exteroceptive integration; ACC contributes evaluative control and autonomic set (Wang et al., 2019). **Critically,** we distinguish two separable components of the self-prior:

1. **Interoceptive signal transmission** (AIC-mediated): Can the bodily signal reach cortex? Measurable via HEP amplitude and interoceptive accuracy tasks.
2. **Self-binding/ownership attribution** (AIC–PCC/TPJ-mediated): Does the signal feel like **mine**? Measurable via AIC–PCC connectivity and ownership-illusion paradigms.

This distinction is crucial for understanding depersonalization: **Signals can be detected (component 1 intact) while ownership fails (component 2 disrupted)** (Medford et al., 2016). We predict DPD shows normal or even heightened HEP amplitude but reduced AIC–PCC functional connectivity, especially during interoceptive tasks. Posterior parietal/TPJ and posterior cingulate (PCC) provide the ownership tag that binds interoceptive content to the first-person perspective (Salomon et al., 2016).

Cardiac/respiratory entrainment and phase landscape. The self-prior is **not static** but oscillates rhythmically with bodily cycles, creating a body-synchronized variance schedule. Perception, attention, and memory encoding fluctuate with cardiac (~1 Hz) and respiratory (~0.2–0.3 Hz) phase, suggesting that $\boldsymbol{\lambda(t)}$ **itself is rhythmic**.

We predict a **continuous phase landscape** of denoising efficiency:

$$\mathcal{D}\big(\varphi_{\text{cardiac}}, \varphi_{\text{resp}}, \varphi_{\text{theta}}\big) = \mathcal{D}_0 \left[1 + \sum_i w_i \cos(\varphi_i - \varphi_i^*)\right] \prod_{j<k} \big[1 + c_{jk} \cos\big(\varphi_j - \varphi_k\big)\big] \tag{4}$$

where $\mathcal{D}$ is denoising efficiency, φ_i^* are optimal individual phases, w_i are phase weights, and c_{jk} capture phase–phase interactions.

Illustrative future targets:

- **Primary peaks:** The location, magnitude, and even the number of phase optima must be estimated rather than fixed a priori. Candidate cardiac, respiratory, and theta combinations provide a preregisterable search space, not numerical NDT predictions.
- **Secondary structure:** Additional peaks/troughs at cardiac–respiratory harmonic combinations (2:1, 3:1 ratios).
- **Empirical validation:** Measure entropy $e(t)$, recall performance, and subjective clarity as function of 3D phase coordinates $\left(\varphi_{\text{cardiac}}, \varphi_{\text{resp}}, \varphi_{\text{theta}}\right)$ during encoding.

Practical operationalization. Since < 100 ms phase precision exceeds current phenomenological methods, we test via (i) phase-triggered stimulus presentation with neural/behavioral readout, (ii) post-hoc phase-binning of continuous encoding data, and (iii) phase-shuffling controls to verify that the observed structure is not random.

Pathological phase relationships:

- **Anxiety:** Over-rigid cardiac–neural coupling (pathologically locked $\lambda(t)$), creating context-insensitive hyper-anchoring. Predict: elevated HEP–theta phase-locking across contexts, even when this is contextually inappropriate.
- **Depersonalization:** Absent or random cardiac/respiratory–neural coupling (decorrelated $\lambda(t)$). Predict: no systematic phase effects on perception, flat HEP–context interactions.

These phase-coupling patterns may be measurable and manipulable in future biofeedback studies, but that possibility is an exploratory research hypothesis rather than a therapeutic claim.

Section takeaway (EAP for this manuscript). Within the present paper, EAP's role is primarily to (i) formalize a self-prior term F_{self} and its coupling to interoceptive signals, and (ii) sketch how this could, in principle, modulate denoising schedules and MNPS geometry. Detailed phase landscapes, SPI-style composite indices, and therapeutic applications are treated as Tier-2, future-work directions rather than requirements for the initial validation of NDT's core sleep and clinical geometry predictions.

S3.5 Exploratory phenomenology, pathology, and training

Mirror failures. Under-anchoring (DPD): Denoising proceeds but fails to bind to self (high Ω yet low ownership), associated with low/flat $\lambda(t)$ or weak AIC–PCC coupling. **Over-anchoring** (panic/anxiety): Bodily noise overweighted, forcing threat-consistent reconstructions (exaggerated, context-insensitive $\lambda(t)$). EAP thus predicts **topologically distinct** failures in MNPS (low recurrence and unstable hubs in DPD; over-recurrence in anxiety during anticipatory phases).

Trainability. Interoceptive training (breath/heartbeat attention) may recalibrate EAP: increase AIC–ACC coupling, restore context-sensitive $\lambda(t)$ modulation, and alter component readouts linked to curvature/variance **flexibility** (anchored exploration). These training and pharmacology ideas are exploratory roadmap claims, not clinical recommendations.

S3.6 Measurement plan and composite index

Self-prior index (SPI). We propose a composite measure combining multiple EAP signatures:

1. **Signal strength:** HEP amplitude and latency (baseline interoceptive precision).
2. **Context sensitivity:** HEP modulation by task/context (for example, HEP difference between safe vs. threat contexts, or cardio-visual synchrony effects).
3. **Anatomical substrate:** AIC/ACC activation during interoceptive tasks (fMRI/PET).
4. **Binding network:** AIC–PCC/TPJ functional connectivity (distinguishes signal from ownership).
5. **Phase coupling:** Cardiac/respiratory–neural phase coherence and its task-dependent modulation.

Predicted covariations. SPI should correlate with (i) cycle-averaged coherence $|(\Omega)$ during active denoising (tasks, encoding), (ii) Atlas component profiles linked to flexible geometry (see **Supplementary Methods S1**), and (iii) subjective presence/embodiment ratings. After interoceptive training or successful therapy for DPD/anxiety, SPI should normalize— specifically, **context sensitivity** (component 2) and **binding connectivity** (component 4) should improve more than raw signal strength (component 1).

Estimation pipeline for $\boldsymbol{\lambda(t)}$ with artifact control. We standardize EAP estimation as follows: (i) extract R-peaks and respiratory phase; (ii) compute HEP using robust averaging with ECG/EOG artifact regression and cardioballistic artifact suppression (template subtraction); (iii) estimate instantaneous $\lambda_{\text{signal}}(t) \propto A_{\text{HEP}}$ and $\lambda_{\text{binding}}(t)$ via AIC–PCC functional connectivity; (iv) derive phase-aligned coupling terms for Eq. 2; (v) validate with phase-shuffled surrogates and cross-condition controls (safe vs. threat) (Coll et al., 2021; Park & Blanke, 2019).

Falsifiable contrasts. If EAP is correct: (a) HEP/context interactions predict presence/ownership beyond raw HEP; (b) DPD shows normal interoceptive **detection** but reduced insula–PCC coupling and weak $\lambda(t)$

modulation; (c) phase-locked sampling benefits vanish when cardiac/respiratory phases are randomized or misaligned.

Status ceiling

EAP remains Tier-2. Current companion evidence supports bounded, condition-level autonomic/MNPS co-modulation in wake and carrier-specific (SO-trough-dominant) autonomic phase-locking in sleep. It does **not** support causal body-to-manifold gating, phase-resolved variance entrainment (Eq. 2 remains an untested formal target), HEP-based binding, or clinical translation. The hypoglycemia contrast documents a bounded metabolic MNPS dissociation, not an EAP metabolic gate.

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